Portfolio Optimization and Parameter Uncertainty

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Abstract

Portfolio optimization has a mixed reputation among investment managers, with some being so skeptical that they believe it is almost useless due to the inherent parameter uncertainty. It is undeniable that portfolio optimization problems are sensitive to parameter estimates, especially the expected returns that are arguably also the hardest parameters to estimate. However, most practitioners still attempt to build mean-risk optimal portfolios, albeit in implicit ways. Resampled optimization is a popular mathematical heuristic to tackle the parameter uncertainty issue. It computes optimal portfolios using sampled parameter estimates and calculates a simple average of the portfolio exposures across samples. The unsatisfactory aspect of the resampled approach is that there is no mathematical justification for using the average of portfolio exposures, it just works well in practice. This article provides perspectives for understanding the resampling approach by analyzing the portfolio exposure estimation process from a bias-variance trade-off. We show that the traditional resampled optimization corresponds to a naive version of stacked generalization. Finally, we introduce a stacked generalization approach that can be used to handle both parameter uncertainty and combine optimization methods in full generality. We coin the new method Exposure Stacking.

Documented Python code that replicates the results of the case study is available in the open-source package fortitudo.tech. More information about the package can be found on https://os.fortitudo.tech.

Keywords: Portfolio optimization, parameter uncertainty, Exposure Stacking, mean-CVaR, tail risk, mean-variance, efficient portfolio, efficient frontier, mean squared error, bias-variance trade-off, stacked generalization, quadratic programming, convex optimization, Python Programming Language.

1 Introduction

Portfolio optimization remains one of the most researched areas of quantitative investment management. While it is well-known that portfolio optimization problems are highly sensitive to parameter estimates, or more generally the market model input, most people still attempt to build mean-risk optimal portfolios. Some practitioners are so skeptical that they do not use portfolio optimization methods at all, at least explicitly.

This article argues that although the parameter uncertainty problem is a real and significant issue, there exists practically feasible ways to handle the problem in full generality. We also note that the issues stemming from parameter uncertainty can be mitigated by properly setting up the optimization problem with risk budgets and transaction costs, see Vorobets (2022b) for the formulation of portfolio optimization problems having this form and general risk measures.

We argue that the parameter uncertainty issue can be analyzed from the same perspective as statistical learning models using the bias-variance trade-off. Specifically, we treat the market model or parameters as data and the optimal portfolio exposures as estimates, with the particular mean-risk optimization method being an estimator. From this perspective, we argue that traditional mean-risk portfolio optimization estimators have no bias but a high variance due to their sensitivity to the market model and parameter values. On the other hand, resampled versions of these estimators have some bias but a lower variance.

When we perform resampled portfolio optimization, we are purposefully introducing bias into the mean-risk model while decreasing its variance. The issue with the original resampled optimization introduced by Michaud and Michaud (1998) is that we have no guarantee that the resulting trade-off between bias and variance is optimal when we take a simple average of the portfolio exposures. Critiques of the resampling approach are also related to the lack of mathematically sound arguments for using the procedure, which this article provides stacked generalization solutions for.

Stacked generalization, see Wolpert (1992) and Breiman (1996), has recently received increased attention due to its versatility and potential for improving out-of-sample performance. For the parameter uncertainty issue, we apply the stacked generalization ideas from a slightly different perspective, while our method still works in the traditional sense to combine the predictions of different portfolio optimization models. While our case study finds that Exposure Stacking is no worse and sometimes better than the traditional resampled approach, the risk-adjusted performance gains must still be assessed by extensive practical use of the method. Conveniently, Exposure Stacking can be formulated as a quadratic programming problem, making it easy to explore and solve in a fast and stable way.

Several different suggestions have been proposed for handling the parameter sensitivity issue and the resulting concentrated portfolios. We will purposefully not survey these and focus on the resampled approach, which seems to be the most frequently used method among investment practitioners, and which we think is the most versatile. The rest of this article is organized as follows: Section 2 presents the framework, Section 3 is a case study that analyzes which parameters contribute the most to portfolio sensitivity in the mean-variance setting and compares Exposure Stacking to traditional resampled optimization, while Section 4 is a conclusion.

2 The Exposure Stacking method

Some variant of the resampling approach introduced by Michaud and Michaud (1998) is often applied in practice as a heuristic way to handle parameter uncertainty in portfolio optimization. The issue with this approach is that there are no mathematical justification for computing a simple average of the resampled

portfolio exposures. The new Exposure Stacking method introduced in this article selects the individual sample weights by minimizing a stacking objective, which can be solved in a fast and stable way using quadratic programming.

To understand the new method, it is important to view the portfolio optimization problem from the right perspective. Hence, we start by defining it here. We consider a risk-adjusted return objective f, which is a function of the market model (R, p) with $R \in \mathbb{R}^{S \times I}$ being a matrix of joint market scenarios and $p \in \mathbb{R}^S$ being an associated scenario probability vector, see Vorobets (2021) for a further description of the market representation. Additionally, f is a function of the optimization constraints $\mathcal{E}$ and the portfolio exposures $e \in \mathbb{R}^I$. First, consider the case where f determines the expected return, while constraints on the risk are included in $\mathcal{E}$. The optimal portfolio exposures $e^\star$ are then given by

$$e^\star = \underset{e}{\operatorname{argmax}}\ f\left(R, p, \mathcal{E}, e\right).$$

It is also possible to work from the perspective of minimizing risk with a return target. In such cases, the optimal portfolio exposures are given by

$$e^\star = \underset{e}{\operatorname{argmin}}\ g\left(R, p, \mathcal{E}, e\right),$$

where g is a function that determines the risk, while the return target is included in the constraints $\mathcal{E}$. We note that the two perspectives solve equivalent risk-adjusted return problems.

The risk-adjusted objective can be defined for any portfolio risk measure, for example, variance or CVaR. In practice, we recommend using CVaR given empirical return distributions are usually very far from being normally or elliptically distributed, see Vorobets (2022b) for further details. When the investment universe contains options, the linear and constant dependency assumption inherent to mean-variance will also obviously be violated, see Vorobets (2022a) for a comparison between CVaR and variance optimization with options.

To explore resampling from a bias-variance perspective, we define the optimal exposure for sample $b \in \{1, 2, \ldots, B\}$ by

$$e_b^\star = \underset{e}{\operatorname{argmax}}\ f\left(R_b, p_b, \mathcal{E}, e\right).$$

Note that in this definition, we are optimizing over samples (R_b, p_b) of the market model. For the mean-variance objective, this is simply estimates of mean vectors and covariance matrices. For the mean-CVaR objective, this can be entirely new joint market scenarios R_b and associated probability vectors p_b, with p_b possibly generated using Sequential Entropy Pooling (SeqEP) as described in Vorobets (2021), Vorobets (2023), and Vorobets (2024).

Associated with each sample exposure $e_b^\star$, we have the opportunity of selecting sample weights $w_b \in [0, 1]$ with $\sum_{b=1}^B w_b = 1$, which we use to estimate the final weighted resampled exposures vector

$$e_w^\star = \sum_{b=1}^{B} w_b e_b^\star. \tag{1}$$

The traditional resampled portfolio estimator introduced by Michaud and Michaud (1998) corresponds to $w_b = \frac{1}{B}$ for all $b \in \{1, 2, \ldots, B\}$.

How do we analyze the risk-adjusted objective from a bias-variance perspective? We must first understand in which sense the traditional portfolio optimization methods such as mean-variance and mean-CVaR are unbiased. We call a mean-risk estimator unbiased if it correctly estimates $e^\star$, i.e., if it correctly estimates

the optimal portfolio exposures given the market model (R, p). Clearly, the resampled estimator (1) is not unbiased in the general case.

It is well-known that mean-risk portfolio optimization is highly sensitive to the market model and parameter values, meaning that it has a high variance. When we perform resampled optimization, we are purposefully introducing bias to reduce the variance of the final optimal exposures estimate. If we weigh the sample estimates equally as originally suggested, we are just hoping that the increase in bias will reduce the variance more. Although this is also what people experience in practice, we do not have any guarantees that the bias-variance trade-off is minimized by an equally weighted average.

Direct minimization of the vector mean squared error

$$MSE\left(e_w^\star\right) = \mathbb{E}\left[\left\|\sum_{b=1}^{B} w_b e_b^\star - e^\star\right\|_2^2\right]$$

of the resampled estimator (1) is not interesting given that the solution will simply try to find a linear combination of the vectors $e_b^\star$ which is as close as possible to $e^\star$ with respect to the Euclidean norm $\|\cdot\|_2$. Instead, we focus on multivariate stacking objectives having a form similar to the vector mean squared error.

Stacked generalization, see Wolpert (1992) and Breiman (1996), has received increased attention in recent years due to its versatility and potential for improving out-of-sample performance. Hence, we analyze the resampled portfolio optimization problem from this perspective. A natural, albeit slightly naive, starting point for the objective function is to ensure that the Euclidean norm of the difference between $e_w^\star$ and each $e_b^\star$ is, on average, as small as possible, i.e., solve the problem

$$w^\star = \underset{w}{\operatorname{argmin}} \frac{1}{B}\sum_{k=1}^{B}\left\|e_k^\star - \sum_{b=1}^{B} w_b e_b^\star\right\|_2^2. \tag{2}$$

However, as we show in Appendix A, the optimal solution to (2) always yields a vector of equal sample weights $w_b^\star = \frac{1}{B}$ for all $b \in \{1, 2, \ldots, B\}$, corresponding to the traditional resampling method.

Instead of (2), we define an objective function based on the ideas of stacked regression and cross-validation. Let $\mathcal{B} = \{1, 2, \ldots, B\}$ be the set of sample indices and suppose that we partition $\mathcal{B}$ into L nonempty sets $\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_L$ for some $L \in \{2, 3, \ldots, B\}$. For any choice of $l \in \{1, 2, \ldots, L\}$, we consider the set of exposure vectors $\{e_k \mid k \in \mathcal{K}_l\}$ as a validation set, find sample weights w_b for the remaining $B - |\mathcal{K}_l|$ exposure vectors, and calculate the average difference ε_l between the weighted exposure and the validation set exposures, i.e., we compute

$$\varepsilon_l = \frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}\left\|e_k^\star - \sum_{b\notin\mathcal{K}_l} w_b e_b^\star\right\|_2^2.$$

If we repeat the analysis using each of the sets $\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_L$ as our validation set indices, we can compute the L-fold cross-validation estimate as the average $\varepsilon = \frac{1}{L}\sum_{l=1}^{L}\varepsilon_l$, see James et al. (2023). Thus, instead of using the naive objective function (2), we wish to find the vector of sample weights $w^\star$ that minimizes the cross-validation estimate ε, i.e., solve the problem

$$w^\star = \underset{w}{\operatorname{argmin}} \frac{1}{L}\sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}\left\|e_k^\star - \sum_{b\notin\mathcal{K}_l} w_b e_b^\star\right\|_2^2\right). \tag{3}$$

We show in Appendix B that (3) can be formulated as a quadratic programming problem and therefore solved in a fast and stable way. Note that (3) allows us to stack exposures both across different market model samples (R_b, p_b) to incorporate parameter uncertainty as well as different efficient portfolio estimators. For example, one could combine mean-variance, mean-CVaR, or even mean-CVaR with different CVaR levels such as 90%, 95%, and 97.5%. We coin all approaches that solve the problem in (3) Exposure Stacking. Finally, we note that (3) can be used more generally to combine the joint predictions of multiple models in multivariate regression problems.

2.1 Perspectives on risk and return alignment

In the original resampled optimization, portfolios are grouped according to their index on the efficient frontier. This somehow aligns the risk of the sample portfolios $e_b^\star$ and avoids potential issues with significant drift in the risk and return for the final portfolio estimate $e_w^\star$. To our knowledge, there are no recommendations for how to handle resampled portfolio optimization for single portfolios. If we introduce uncertainty only into the means, which we show in Section 3.1 below affects portfolio optimization the most, we can of course optimize portfolios with the same level of risk to avoid significant drift in the risk level of $e_w^\star$. If we introduce uncertainty into both risk and expected return, a suggestion is to find a reference portfolio, compute its risk and return for the different market model samples (R_b, p_b) and use that as targets in the B resampled optimizations, i.e., optimize with the same target return or target risk as the reference portfolio.

3 Case study

In this case study, we optimize with respect to portfolio variance to build intuition and because it is fast and easy, but we recommend that you use CVaR for investment management in practice with arguments given by Vorobets (2022b). We note that there might be a patent related to the traditional resampled optimization introduced by Michaud and Michaud (1998) when the average is computed over samples where uncertainty has been introduced into both the mean vectors and covariance matrices. Hence, we purposefully do not perform these computation in this article or the accompanied code.[1]

As we show in Section 3.1 below, uncertainty in the expected returns generally affects the optimal exposures $e_b^\star$ more than uncertainty in variances and correlations. Therefore, when comparing equal weight resampling to Exposure Stacking in Section 3.2, we will do so on the basis of the resampled frontier with only mean uncertainty. We have no reason to believe that our conclusions will change if uncertainty is introduced into all parameters of the mean-variance model. We encourage the interested reader to adapt the accompanied code and introduce market model uncertainty however they wish to assess for themselves.

3.1 Analysis of parameter importance

We analyze which parameters affect the efficient allocations the most by introducing uncertainty into means, variances, and correlations. We use the same parameters as in Vorobets (2022b) to generate normally distributed samples with $N \in \{10, 50, 100\}$ observations and compute $B = 1,000$ efficient frontiers. Figure 1, Figure 2, and Figure 3 show these efficient frontiers. In the figures, red, green, and black represent portfolios with respectively mean, variance, and correlation uncertainty. We clearly see that mean uncertainty affects the efficient frontiers the most and encourage readers to examine the accompanied code for additional details.

[1] https://github.com/fortitudo-tech/fortitudo.tech/blob/main/examples/9_PortfolioOpt_ParamUncertainty.ipynb

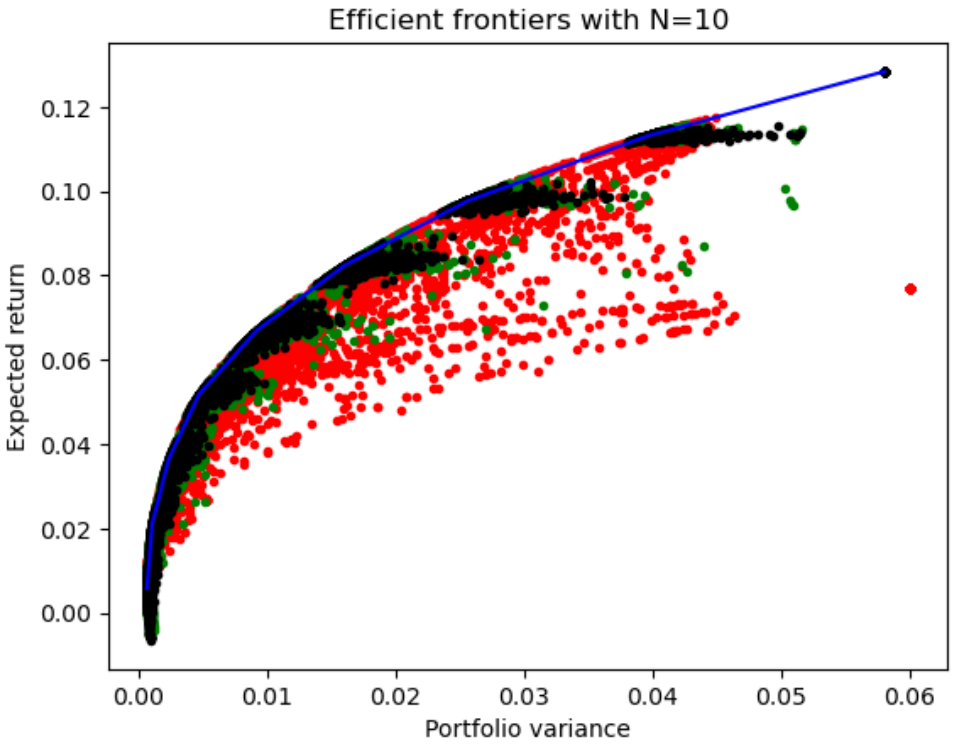


Figure 1: Efficient frontiers with $N = 10$ estimation observations.

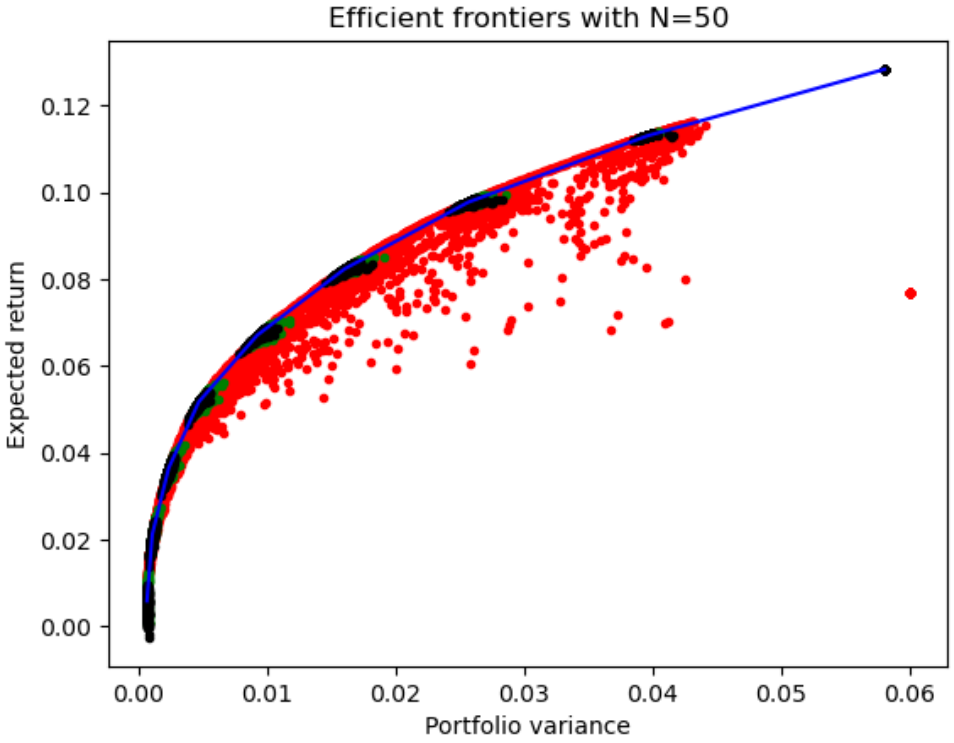


Figure 2: Efficient frontiers with $N = 50$ estimation observations.

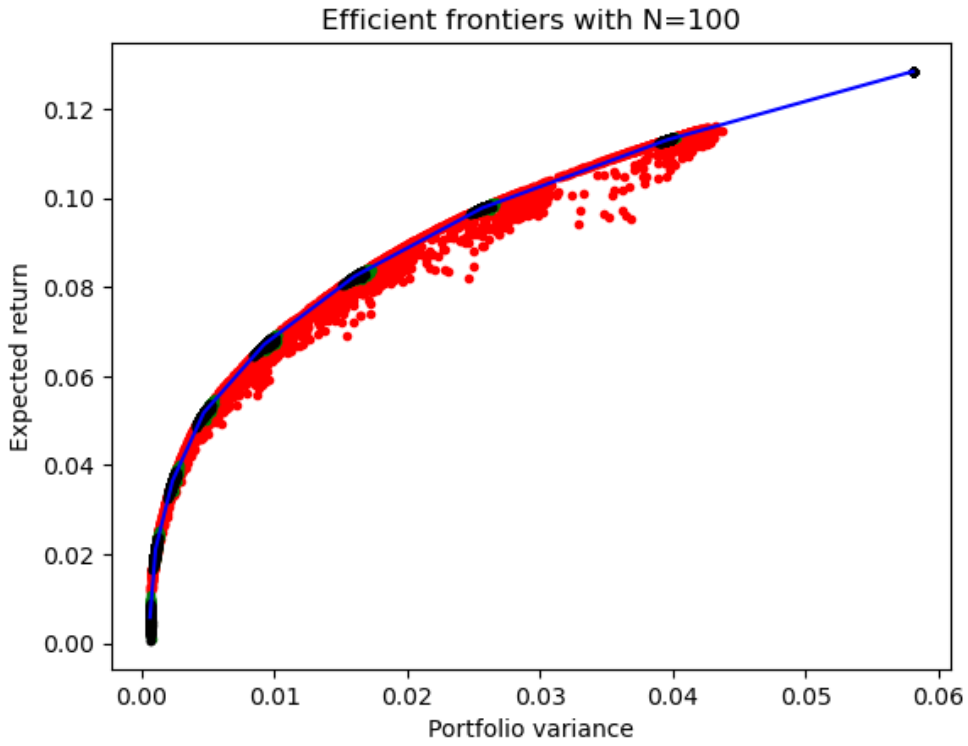


Figure 3: Efficient frontiers with $N = 100$ estimation observations.

3.2 Comparing equal weight resampling to Exposure Stacking

To evaluate Exposure Stacking and compare it to the traditional resampling method, $B = 1,000$ efficient frontiers with mean uncertainty are computed using $N = 100$ estimation observations. Each of these efficient frontiers consists of nine portfolios, and we examine the middle portfolio. We compute the L-fold Exposure Stacking portfolio for $L \in \{2, 5, B\}$ by dividing the $1,000$ sample exposures $e_b^\star$ into L sets of equal size and solving the quadratic programming problem (4) from Appendix B. We note that when performing Exposure Stacking, the results may depend on how the sample portfolios $e_b^\star$ are partitioned. Since we only use one mean-risk estimator for all samples and each sample is i.i.d., we simply partition the sample exposures $e_b^\star$ in the sampled order.

To compare the performance of L-fold Exposure Stacking with equal weights resampling, we simulate another $1,000$ out-of-sample means and covariances and use those to compute the risk and return of the equal weight resampled exposure, the exposure from the efficient frontier, and each of the L-fold Exposure Stacking results. The distribution of the risk-adjusted returns can be seen in Figure 4, which shows that all resampling methods have better out-of-sample risk-adjusted returns than the frontier portfolio. It is worth noting that the traditional resampling and the B-fold Exposure Stacking have practically identical distributions in this case, and that the difference between 5-fold and B-fold Exposure Stacking is negligible. On the other hand, the 2-fold Exposure Stacking has a better risk-adjusted return than the other methods. As can be seen in Figure 5, the traditional resampling and 2-fold Exposure Stacking have similar out-of-sample returns, but 2-fold Exposure Stacking has a lower risk, which results in the improved risk-adjusted return. In comparison, 5-fold Exposure Stacking has higher returns but also higher risk, and B-fold almost coincides with the traditional resampling method in this case study.

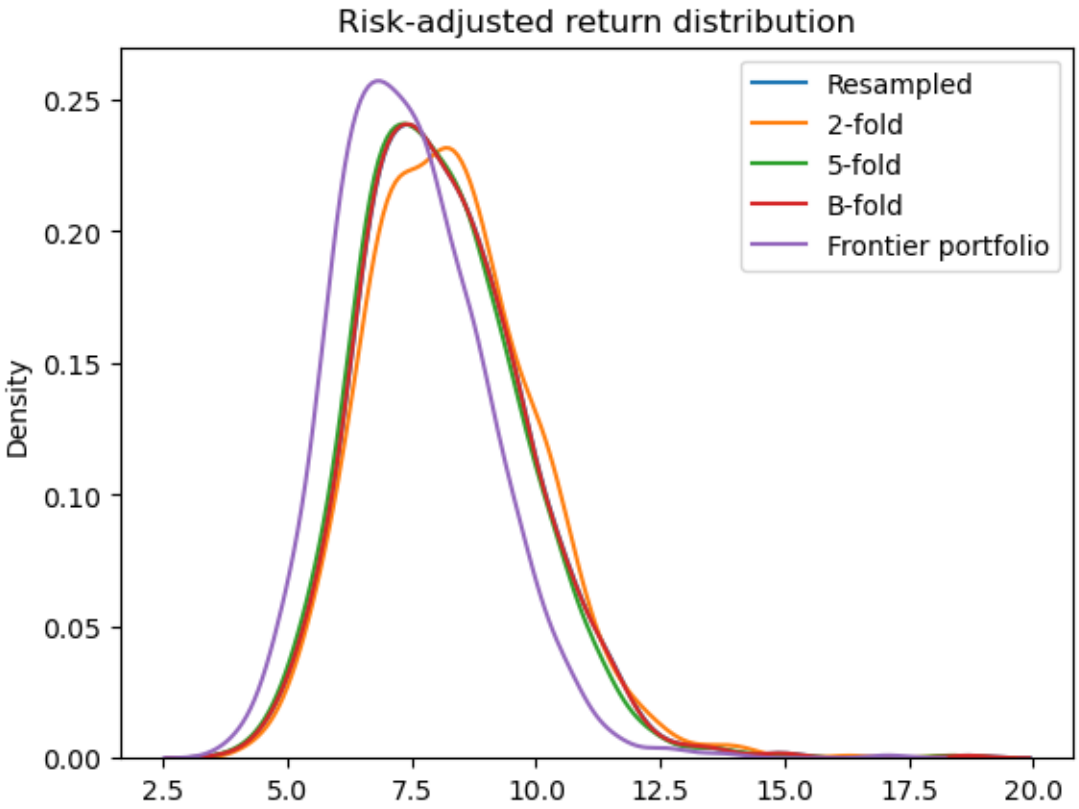


Figure 4: Distribution of the risk-adjusted returns.

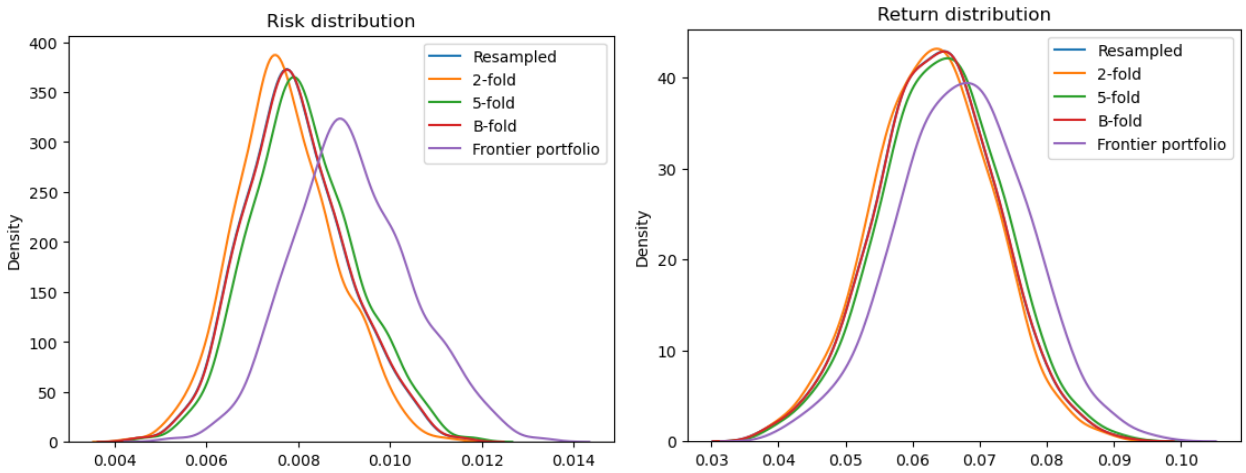


Figure 5: Out-of-sample risk and return distributions.

It is important to underline that this is a relatively small and simple case study, where each sample exposure vector $e_b^\star$ is computed using just mean-variance optimization with basic long-only constraints. Thus, it is too early to make definite conclusions on the magnitude of the risk-adjusted improvements due to Exposure Stacking compared to the traditional resampling method. However, we find that Exposure Stacking performs no worse than the naive approach and has potential for better out-of-sample performance.

To further assess the risk-adjusted gains stemming from Exposure Stacking, extensive practical use is needed. We encourage readers to share their findings and send us feedback.

3.3 Other experiments

We have experimented with applying stacking to the portfolio return, portfolio risk, and portfolio risk-adjusted return. Our limited experience suggests that this gives results that deviate quite significantly from the average portfolio and cause a drift in the overall risk and return level of the final estimate $e_w^\star$. We could of course have added some constraints on how much the expected portfolio return is allowed to deviate from the average. This is also possible for risk, but more challenging. We encourage the interested reader to experiment with these approaches as well as various ways of stacking used in combination with resampled parameters or market model uncertainty. When the portfolios $e_b^\star$ are optimized with the same constraints, our experience is that Exposure Stacking does not cause a significant drift in the risk and return levels.

4 Conclusion

This article introduces novel perspectives and results for portfolio optimization with parameter uncertainty, having a particular focus on resampled optimization. We introduce a new method for computing the sample weights of the resampled portfolios, which we coin Exposure Stacking. The new approach allows us to handle fully general market model uncertainty and combine multiple portfolio optimization methods having different risk measures and hyperparameters. The case study shows that Exposure Stacking has the potential for better out-of-sample performance than the traditional resampling method. To further assess the practical risk-adjusted gains, extensive use of the new Exposure Stacking method is required.

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Appendix A: Naive stacking

Let $w = (w_1, w_2, \ldots, w_B)^T$ be the vector of sample weights, and let $M \in \mathbb{R}^{B \times I}$ denote the matrix where row b consists of the transposed exposure vector $e_b^{\star T}$ for all $b \in \{1, 2, \ldots, B\}$, i.e.,

$$M = \begin{bmatrix} e_1^{\star T} \\ \vdots \\ e_B^{\star T} \end{bmatrix} \in \mathbb{R}^{B \times I}.$$

We note that

$$M^T w = \begin{bmatrix} e_{11}^\star & \cdots & e_{B1}^\star \\ \vdots & \ddots & \vdots \\ e_{1I}^\star & \cdots & e_{BI}^\star \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_B \end{bmatrix} = \begin{bmatrix} \sum_{b=1}^B w_b e_{b1}^\star \\ \vdots \\ \sum_{b=1}^B w_b e_{bI}^\star \end{bmatrix} = \sum_{b=1}^B w_b \begin{bmatrix} e_{b1}^\star \\ \vdots \\ e_{bI}^\star \end{bmatrix} = \sum_{b=1}^B w_b e_b^\star,$$

where $e_{bi}^\star$ denotes entry i of $e_b^\star$ for all $i \in \{1, 2, \ldots, I\}$ and $b \in \{1, 2, \ldots, B\}$.

Let $f(w)$ denote the objective function of the naive stacking problem (2), excluding the constant $\frac{1}{B}$ as it will not affect the solution. Note that $f(w)$ can be rewritten as

$$\begin{aligned}
f(w) &= \sum_{k=1}^B \left\| e_k^\star - \sum_{b=1}^B w_b e_b^\star \right\|_2^2 \\
&= \sum_{k=1}^B \left(e_k^\star - \sum_{b=1}^B w_b e_b^\star \right)^T \left(e_k^\star - \sum_{b=1}^B w_b e_b^\star \right) \\
&= \sum_{k=1}^B \left(e_k^\star - M^T w \right)^T \left(e_k^\star - M^T w \right) \\
&= \sum_{k=1}^B \left(e_k^{\star T} - w^T M \right) \left(e_k^\star - M^T w \right) \\
&= \sum_{k=1}^B e_k^{\star T} e_k^\star - e_k^{\star T} M^T w - w^T M e_k^\star + w^T M M^T w \\
&= \sum_{k=1}^B w^T M M^T w - 2 e_k^{\star T} M^T w + e_k^{\star T} e_k^\star \\
&= B w^T M M^T w - 2 \left(\sum_{k=1}^B e_k^{\star T} \right) M^T w + \sum_{k=1}^B e_k^{\star T} e_k^\star \\
&= \frac{1}{2} w^T P w + q^T w + c,
\end{aligned}$$

where $P \in \mathbb{R}^{B \times B}$, $q \in \mathbb{R}^B$, and $c \in \mathbb{R}$ are given by

$$\begin{aligned} P &= 2BMM^T, \\ q &= -2M \sum_{k=1}^{B} e_k^\star, \\ c &= \sum_{k=1}^{B} e_k^{\star T} e_k^\star. \end{aligned}$$

Since c does not depend on w and therefore does not affect the solution, the naive stacking problem (2) can be expressed as the quadratic programming problem

$$\begin{aligned} \underset{w}{\text{argmin}} \ & \frac{1}{2} w^T P w + q^T w \\ \text{s.t.} \ & \sum_{b=1}^{B} w_b = 1, \\ & w \geq 0, \end{aligned}$$

where $w \geq 0$ denotes entrywise inequalities.

Let ν and $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_B)$ denote the Lagrange multipliers associated with, respectively, the equality and inequality constraints. Since quadratic programming problems are known to be convex, any feasible solution $w^\star$ and associated Lagrange multipliers that satisfy the Karush-Kuhn-Tucker (KKT) conditions yields an optimal solution, see Boyd and Vandenberghe (2004).

The Lagrangian of the naive stacking problem (2) is given by

$$\mathcal{L}(w, \lambda, \nu) = f(w) - \sum_{b=1}^{B} \lambda_b w_b + \nu \left(\sum_{b=1}^{B} w_b - 1 \right).$$

We further note that we can rewrite $f(w)$ as

$$\begin{aligned} f(w) &= \sum_{k=1}^{B} \left\| e_k^\star - \sum_{b=1}^{B} w_b e_b^\star \right\|_2^2 \\ &= \sum_{k=1}^{B} \left(\sum_{i=1}^{I} \left(e_{ki}^\star - \sum_{b=1}^{B} w_b e_{bi}^\star \right)^2 \right) \\ &= \sum_{k=1}^{B} \left(\sum_{i=1}^{I} \left(e_{ki}^{\star 2} + \left(\sum_{b=1}^{B} w_b e_{bi}^\star \right)^2 - 2 e_{ki}^\star \sum_{b=1}^{B} w_b e_{bi}^\star \right) \right), \end{aligned}$$

where $e_{bi}^\star$ denotes entry i of $e_b^\star$ as before.

For any $l \in \{1, 2, \ldots, B\}$, the partial derivative of f with respect to w_l is

$$\begin{aligned}
\frac{\partial f(w)}{\partial w_l} &= \sum_{k=1}^{B} \left(\sum_{i=1}^{I} \left(2 \sum_{b=1}^{B} w_b e_{bi}^\star e_{li}^\star - 2 e_{ki}^\star e_{li}^\star \right) \right) \\
&= 2 \sum_{i=1}^{I} \left(\sum_{k=1}^{B} \left(\sum_{b=1}^{B} w_b e_{bi}^\star e_{li}^\star - e_{ki}^\star e_{li}^\star \right) \right) \\
&= 2 \sum_{i=1}^{I} \left(B \sum_{b=1}^{B} w_b e_{bi}^\star e_{li}^\star - \sum_{k=1}^{B} e_{ki}^\star e_{li}^\star \right).
\end{aligned}$$

Hence, the partial derivative of the Lagrangian $\mathcal{L}(w, \lambda, \nu)$ with respect to w_l is given by

$$\frac{\partial \mathcal{L}}{\partial w_l} = 2 \sum_{i=1}^{I} \left(B \sum_{b=1}^{B} w_b e_{bi}^\star e_{li}^\star - \sum_{k=1}^{B} e_{ki}^\star e_{li}^\star \right) - \lambda_l + \nu.$$

Thus, the KKT conditions for the primal optimal $w^\star$ and dual optimal $(\lambda^\star, \nu^\star)$ are given by

$$\begin{aligned}
\sum_{b=1}^{B} w_b^\star &= 1, \\
w_b^\star &\geq 0 \qquad \forall b \in \{1, 2, \ldots, B\}, \\
\lambda_b^\star &\geq 0 \qquad \forall b \in \{1, 2, \ldots, B\}, \\
\lambda_b^\star w_b^\star &= 0 \qquad \forall b \in \{1, 2, \ldots, B\}, \\
2 \sum_{i=1}^{I} \left(B \sum_{b=1}^{B} w_b e_{bi}^\star e_{li}^\star - \sum_{k=1}^{B} e_{ki}^\star e_{li}^\star \right) - \lambda_l^\star + \nu^\star &= 0 \qquad \forall l \in \{1, 2, \ldots, B\},
\end{aligned}$$

Let $w^\star = \left(\frac{1}{B}, \frac{1}{B}, \ldots, \frac{1}{B}\right)^T \in \mathbb{R}^B$ be the vector of equal sample weights, and let $\nu^\star = 0$ and $\lambda_b^\star = 0$ for all $b \in \{1, 2, \ldots, B\}$. Then clearly the weights sum to one, $w_b^\star, \lambda_b^\star \geq 0$ and $\lambda_b^\star w_b^\star = 0$ for all $b \in \{1, 2, \ldots, B\}$, which fulfills the first four conditions. To see that the last condition also holds, we note that for any $l \in \{1, 2, \ldots, B\}$ we have

$$\begin{aligned}
2 \sum_{i=1}^{I} \left(B \sum_{b=1}^{B} w_b e_{bi}^\star e_{li}^\star - \sum_{k=1}^{B} e_{ki}^\star e_{li}^\star \right) - \lambda_l^\star + \nu^\star &= 2 \sum_{i=1}^{I} \left(B \sum_{b=1}^{B} \frac{1}{B} e_{bi}^\star e_{li}^\star - \sum_{k=1}^{B} e_{ki}^\star e_{li}^\star \right) \\
&= 2 \sum_{i=1}^{I} \left(\sum_{b=1}^{B} e_{bi}^\star e_{li}^\star - \sum_{k=1}^{B} e_{ki}^\star e_{li}^\star \right) \\
&= 0,
\end{aligned}$$

which fulfills the last condition. Since the problem is convex and all KKT conditions are fulfilled, $w^\star = \left(\frac{1}{B}, \frac{1}{B}, \ldots, \frac{1}{B}\right)^T$ is the optimal solution to the naive stacking problem.

Appendix B: Exposure Stacking

Let $\mathcal{B} = \{1, 2, \ldots, B\}$ with $B \geq 2$ denote the index set of the sample portfolios $e_b^\star$. For a fixed $L \in \{2, 3, \ldots B\}$, we can partition $\mathcal{B}$ into L disjoint and nonempty sets $\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_L$, i.e., $\mathcal{K}_l \neq \emptyset$, $\mathcal{K}_l \cap \mathcal{K}_{l'} = \emptyset$ for all $l, l' \in \{1, 2, \ldots L\}$, and

$$\mathcal{B} = \bigcup_{l=1}^{L} \mathcal{K}_l.$$

For each $l \in \{1, 2, \ldots L\}$, we define the matrix $M_l \in \mathbb{R}^{B \times I}$ as the matrix where row b is $e_b^{\star T}$ if $b \notin \mathcal{K}_l$ and the zero vector if $b \in \mathcal{K}_l$. Additionally, let $m_{bi}^{(l)}$ denote the entry in row b and column i of M_l such that

$$m_{bi}^{(l)} = \begin{cases} e_{bi}^\star & \text{if } b \notin \mathcal{K}_l \\ 0 & \text{if } b \in \mathcal{K}_l \end{cases},$$

where $e_{bi}^\star$ is entry i of the exposure vector $e_b^\star$. For the vector of sample weights $w = (w_1, w_2, \ldots, w_B)^T$, we then have

$$\begin{aligned}
M_l^T w &= \begin{bmatrix} m_{11}^{(l)} & \cdots & m_{B1}^{(l)} \\ \vdots & \ddots & \vdots \\ m_{1I}^{(l)} & \cdots & m_{BI}^{(l)} \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_B \end{bmatrix} \\
&= \begin{bmatrix} \sum_{b=1}^{B} w_b m_{b1}^{(l)} \\ \vdots \\ \sum_{b=1}^{B} w_b m_{bI}^{(l)} \end{bmatrix} \\
&= \begin{bmatrix} \sum_{b \in \mathcal{K}_l} w_b m_{b1}^{(l)} + \sum_{b \notin \mathcal{K}_l} w_b m_{b1}^{(l)} \\ \vdots \\ \sum_{b \in \mathcal{K}_l} w_b m_{bI}^{(l)} + \sum_{b \notin \mathcal{K}_l} w_b m_{bI}^{(l)} \end{bmatrix} \\
&= \begin{bmatrix} \sum_{b \notin \mathcal{K}_l} w_b e_{b1}^\star \\ \vdots \\ \sum_{b \notin \mathcal{K}_l} w_b e_{bI}^\star \end{bmatrix} \\
&= \sum_{b \notin \mathcal{K}_l} w_b \begin{bmatrix} e_{b1}^\star \\ \vdots \\ e_{bI}^\star \end{bmatrix} \\
&= \sum_{b \notin \mathcal{K}_l} w_b e_b^\star.
\end{aligned}$$

The L-fold Exposure Stacking objective function (3), excluding the constant $\frac{1}{L}$ which does not affect the solution, can then be written as

$$\begin{aligned}
\sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}\left\|e_k^\star-\sum_{b\notin\mathcal{K}_l}w_b e_b^\star\right\|_2^2\right) &= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}\left(e_k^\star-\sum_{b\notin\mathcal{K}_l}w_b e_b^\star\right)^T\left(e_k^\star-\sum_{b\notin\mathcal{K}_l}w_b e_b^\star\right)\right)\\
&= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}\left(e_k^\star-M_l^Tw\right)^T\left(e_k^\star-M_l^Tw\right)\right)\\
&= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}\left(e_k^{\star T}-w^TM_l\right)\left(e_k^\star-M_l^Tw\right)\right)\\
&= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}e_k^{\star T}e_k^\star-e_k^{\star T}M_l^Tw-w^TM_le_k^\star+w^TM_lM_l^Tw\right)\\
&= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\sum_{k\in\mathcal{K}_l}w^TM_lM_l^Tw-2e_k^{\star T}M_l^Tw+e_k^{\star T}e_k^\star\right)\\
&= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\left(|\mathcal{K}_l|\,w^TM_lM_l^Tw-2\left(\sum_{k\in\mathcal{K}_l}e_k^{\star T}\right)M_l^Tw+\sum_{k\in\mathcal{K}_l}e_k^{\star T}e_k^\star\right)\right)\\
&= \sum_{l=1}^{L}\left(\frac{1}{|\mathcal{K}_l|}\left(\frac{1}{2}w^TP_lw+q_l^Tw+c_l\right)\right)\\
&= \frac{1}{2}w^TPw+q^Tw+c,
\end{aligned}$$

where for each $l\in\{1,2,\ldots,L\}$ we have defined $P_l\in\mathbb{R}^{B\times B}$, $q_l\in\mathbb{R}^B$, and $c_l\in\mathbb{R}$ as

$$\begin{aligned}
P_l &= 2\,|\mathcal{K}_l|\,M_lM_l^T,\\
q_l &= -2M_l\sum_{k\in\mathcal{K}_l}e_k^\star,\\
c_l &= \sum_{k\in\mathcal{K}_l}e_k^{\star T}e_k^\star,
\end{aligned}$$

with $P\in\mathbb{R}^{B\times B}$, $q\in\mathbb{R}^B$, and $c\in\mathbb{R}$ given by

$$\begin{aligned}
P &= \sum_{l=1}^{L}\frac{1}{|\mathcal{K}_l|}P_l,\\
q &= \sum_{l=1}^{L}\frac{1}{|\mathcal{K}_l|}q_l,\\
c &= \sum_{l=1}^{L}\frac{1}{|\mathcal{K}_l|}c_l.
\end{aligned}$$

Since c is constant and independent of w, the L-fold Exposure Stacking problem can be written as the

quadratic programming problem

$$\begin{aligned} \underset{w}{\operatorname{argmin}}\ & \frac{1}{2} w^T P w + q^T w \\ \text{s.t.}\ & \sum_{b=1}^{B} w_b = 1 \\ & w \geq 0, \end{aligned}$$

where $w \geq 0$ denotes entrywise inequalities.

Lastly, we note that in the special case where all validation sets have the same number of samples, i.e., $|\mathcal{K}_l| = |\mathcal{K}_{l'}|$ for all $l, l' \in \{1, 2, \dots L\}$, we have

$$\begin{aligned} P &= \frac{1}{|\mathcal{K}_1|} \sum_{l=1}^{L} P_l, \\ q &= \frac{1}{|\mathcal{K}_1|} \sum_{l=1}^{L} q_l, \end{aligned}$$

which means that scaling the objective function with the constant $\frac{1}{|\mathcal{K}_1|}$ is not strictly necessary, and we can instead solve the problem

$$\begin{aligned} \underset{w}{\operatorname{argmin}}\ & \frac{1}{2} w^T \hat{P} w + \hat{q}^T w \\ \text{s.t.}\ & \sum_{b=1}^{B} w_b = 1 \\ & w \geq 0, \end{aligned} \tag{4}$$

where

$$\begin{aligned} \hat{P} &= \sum_{l=1}^{L} P_l, \\ \hat{q} &= \sum_{l=1}^{L} q_l. \end{aligned}$$